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Visualizing High Dimensional and Big Data

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Abstract

The amount of data created by people, machines and corporations around the world is growing every year. Thanks to innovations such as the Internet of Things, this trend will continue, giving rise to the creation of Big Data. Data visualization leverages principles of visual psychology to help stakeholders identify patterns, trends and correlations that might go undetected in text-based or spreadsheet data. The return on investment (ROI) of big data visualization is well-documented in numerous studies and use cases. However, to achieve ROI from analytics investments, key insights must be uncovered, understood and communicated. Synthesizing huge quantities of data into key insights grows more challenging as data volumes and varieties increase. To address visualization challenges posed by big and high-dimensional data, this paper explores algorithms and techniques that compress the amount of data and/or reduce the number of attributes to be analyzed and visualized. Specifically, this paper examines applying dimensionality reduction and data compression algorithms to reduce attributes, tuples and data points returned to the visualization. By reducing data returned to the visualization, trends, patterns and correlations are easier to view and visualization tool performance is optimized.

Keywords: Data Visualization, Dimensionality Reduction, Parallel Processing, Hadoop, Machine Learning, Big Data

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1. Introduction

The amount of data created by people, machines and corporations around the world is growing every year. Thanks to innovations such as the Internet of Things, this trend will continue, giving rise to the creation of Big Data. Big Data is data that are so voluminous and complex that traditional data-processing applications are inadequate to deal with them. As data volumes and varieties grow, processing and consuming the insights generated becomes challenging.

The curse of dimensionality [1] refers to various challenges that arise when analyzing data in high-dimensional spaces that do not occur in low-dimensional settings. High dimensionality coupled with large sample sizes create issues such as heavy computational cost and algorithmic instability. On the front-end, big data can cause a lag in visualization rendering time and make patterns and trends difficult to identify.

Given these challenges, data reduction and compression techniques should be applied to optimize the visualization of big data. Dimensionality reduction algorithms are applied to reduce the number of attributes in the dataset through mathematical models that represent the principal variables. Techniques include feature selection and feature extraction. Feature selection and feature extraction methods are typically applied before the data is fed to machine learning models. Numerosity reduction techniques are applied to reduce the number of data points in the dataset. Data is replaced or estimated by smaller data representations such as parametric models (which store only the model parameters instead of the actual data), and nonparametric models such as clustering, sampling, and the use of histograms.

In addition to lessening data quality challenges, reducing the dimensionality and numerosity of the data improves operational efficiency by reducing computational and storage costs involved with larger data sets [2]. Fewer dimensions also aid in understanding models, visualizations and the insights they generate. Finally, dimensionality reduction improves model accuracy [2]. The goal of this paper is to examine the impact of several popular dimensionality and numerosity reduction techniques on data visualization.

2. Data Reduction Literature Review

2.1. Reducing Dimensionality: Linear and Non-Linear Techniques

The idea behind dimensionality reduction is to develop a mathematical model that reduces the complexity of the data while still capturing the essential characteristics of the data [3]. It can be divided into linear and non-linear methods, referring to the structure of the underlying manifolds or topological spaces that each class can learn. Linear dimensionality reduction algorithms are limited to learning linear manifolds, while nonlinear dimension reduction algorithms can learn more complex manifolds. Still other algorithms have been implemented in both linear and nonlinear variants [4].

There are dozens of algorithms and techniques for dimensionality reduction [5, 6, 7]. Linear methods project the data to a lower dimensional space by a linear transformation. The approach is relatively easy to execute and interpret and fast to compute. While linear methods are computationally efficient and relatively easy to understand, they might miss non-linear structures in the data. Nonlinear methods focus on preserving neighborhood geometry when mapping points to lower dimensional subspaces, and these subspaces may be curved or otherwise non-Euclidean.

Multi-dimensional scaling [8] is a widely-used method that can be used in a non-linear fashion. It attempts to preserve the dissimilarities between high-dimensional data points in the low-dimensional space. Many methods have been proposed [9] to accelerate its computation. Other methods, such as Isomaps [10] and Local Linear Embedding [11] seek to better preserve local neighborhood structures, and to uncover the intrinsic structure in the data by building a model of manifold connectivity. Unfortunately, these methods are computationally very intensive and heavily rely on the constructed neighborhood graph.

Developments in deep learning have surfaced techniques using auto-encoders [12] and neural networks [13] for dimensionality reduction. An artificial neural network is a mathematical model composed of a large number of neurons that can simulate the structural and functional characteristics of biological neural network. It is a self-adaptive system, which changes the internal structure according to the external input, and is commonly used to model the complex relationship between input and output. Auto-encoder is a complex three-layered neural network [12]. Although achieving comparable performance to traditional methods, deep learning presents a “black-box” approach and lacks a strict theoretical system to support it. Therefore, while impressive performance results using deep learning, we do not know why theoretically [12].

Additionally, there are several dimensionality reduction techniques specifically designed for time series data. These methods specifically exploit the frequential content of the signal and its usual sparseness in the frequency space. The most popular methods are those based on wavelets [14]. Linear and non-linear dimensionality reduction techniques can further be segmented into feature selection and feature extraction methods.

2.2. Reducing Dimensionality: Feature Selection Techniques

Feature selection identifies a subset of the original variables iteratively by finding combinations of subsets and comparing them to the prediction errors [15]. The approach aims to select a small subset of features that minimize redundancy and maximize relevance to the target, such as the class labels in classification [16]. The central premise when using feature selection is that the data contains many features that are either redundant or irrelevant, and can thus be removed without much impairment to relevant information [17]. Redundancy and irrelevancy are two distinct concepts, since a relevant feature may be redundant in the presence of another relevant feature with which it is strongly correlated [18]. Unlike feature extraction, feature selection does not alter the original representation of the data.

According to whether the training set is labelled or not, feature selection algorithms can be categorized into supervised, unsupervised, and semi-supervised techniques [19]. Feature selection techniques can further be categorized into filter models, wrapper models and embedded models. Unsupervised feature selection is a less constrained search problem without class labels, depending on clustering quality measures [20], and can also create valid feature subsets. A comprehensive review of unsupervised feature selection can be found in [21, 22, 23].

Typically, supervised feature selection consists of four basic steps: subset generation, subset evaluation, stopping criterion, and result validation [24]. In the first step, a candidate feature subset will be chosen based on a given search strategy, which is sent, in the second step, to be evaluated according to certain evaluation criterion [16]. The subset that best fits the evaluation criterion will be chosen from all the candidates that have been evaluated after the stopping criterion are met. In the final step, the chosen subset will be validated using domain knowledge or a validation set.

2.2.1.1. Feature Selection Techniques – Filter Strategies

Filter methods select features based on a performance measure, regardless of the data modeling algorithm applied [25]. Only after the best features are found can the modeling algorithms use them. Filter methods can rank individual features or evaluate entire feature subsets. Filters work without taking the classifier into consideration [16]. This makes them very computationally efficient. They are divided into multivariate and univariate methods. Multivariate methods can find relationships among the features, while univariate methods consider each feature separately [26]. A typical filter algorithm consists of two steps [16]. First, filter feature selection methods apply a statistical measure to assign a score to each feature. In the univariate scheme, each feature is ranked independently of the feature space, while the multivariate scheme evaluates features in a batch way [16]. Therefore, the multivariate scheme is naturally capable of handling redundant features [16]. In the second step, the features with highest rankings are chosen to induce classification models [16]. Filter approaches are generally employed where redundancy/irrelevance removal is the aim [27]. The filter model separates feature selection from classifier learning so that the bias of a learning algorithm does not interact with the bias of a feature selection algorithm [16]. It relies on measures of the general characteristics of the training data such as distance, consistency, dependency, information, and correlation [28].

One disadvantage of filter models is that they do not remove multicollinearity. Multicollinearity occurs when one predictor variable in a multiple regression model can be linearly predicted from others with a substantial degree of accuracy. In this situation the coefficient estimates of the multiple regression may change erratically in response to small changes in the model or the data. So, multicollinearity of features must be addressed before training models [29]. Additionally, filter models ignore the effects of the selected feature subset on the performance of the induction algorithm [30]. The optimal feature subset should depend on the specific biases and heuristics of the induction algorithm [28]. Based on this assumption, wrapper models utilize a specific classifier to evaluate the quality of selected features, and offer a simple and powerful way to address the problem of feature selection, regardless of the chosen learning machine [30]. Chi-squared test, Euclidian distance, T-test, CFS-correlation, Markov blanket, FCBFfast correlation, relief, LASSO, Fisher score and information gain based methods [31, 28] are among the feature selection algorithms.

2.2.1.2. Feature Selection Techniques – Wrapper Strategies

Wrapper feature selection algorithms consider the selection of a set of features as a search problem, where feature combinations are prepared, evaluated and compared to other combinations [32, 33]. Wrapper methods embed the model hypothesis search within the feature subset search. In this setup, a search procedure in the space of possible feature subsets is defined, and various subsets of features are generated and evaluated [34]. The evaluation of a specific subset of features is obtained by training and testing a specific classification model, rendering this approach tailored to a specific classification algorithm [35]. To search the space of all feature subsets, a search algorithm is then 'wrapped' around the classification model. However, as the space of feature subsets grows exponentially with the number of features, heuristic search methods are used to guide the search for an optimal subset [36]. A predictive model is used to evaluate a combination of features and assign a score based on model accuracy [37]. The search process may be methodical such as a best-first search, it may stochastic such as a random hill-climbing algorithm, or it may use heuristics, like forward and backward passes to add and remove features [33, 38].

Wrappers tend to perform better in selecting features since they take the model hypothesis into account by training and testing in the feature space [39]. Additionally, wrappers continuously maintain a precise communication between an induction algorithm and the training data. Unlike filters, wrappers can detect feature dependencies [39]. The main disadvantage of wrappers is that they run much slower than filter models because they must repeatedly re-run or call the induction algorithm and every iteration typically demands a cross-validation [40]. As a result, these methods are expensive for high-dimensional data. The feature subsets are also biased towards the modelling algorithm on which they were evaluated (even when using cross-validation) [25]. Thus, for a reliable generalization error estimate, it is necessary that both an independent validation sample and another modelling algorithm are used after the final subset is found. Finally, there exists a higher risk of overfitting than the filter methods [39]. Overfitting occurs when a model is tuned to a particular data set but fails to perform on additional data sets reliably. Wrapper methods include: sequential forward selection, sequential backward selection, randomized hill climbing, genetic algorithms, recursive feature elimination and estimation of distribution algorithms [41].

2.2.1.3. Feature Selection Techniques – Embedded and Hybrid Strategies

Due to the drawbacks of filter and wrapper models, embedded models were proposed to bridge the gap [16]. First, embedded models incorporate the statistical criteria, as filter models do, to select several candidate feature subsets with a given cardinality. Second, they choose the subset with the highest classification accuracy [16]. Thus, embedded models usually achieve both comparable accuracy to the wrapper and comparable efficiency to the filter models [16]. Embedded methods learn which features best contribute to the accuracy of the model while the model is being created [37]. The embedded model performs feature selection in the learning time. So, it achieves model fitting and feature selection simultaneously [42]. Embedded techniques tend to do better computationally than wrappers but they make classifier dependent selections that might not work with any other classifier [39]. Advantages of embedded methods include better use of available data (no need to split training data into a training and validation set) and reaching a solution faster (no retraining a predictor from scratch for each subset) [39]. Examples of embedded selection algorithms are decision trees, random forests, weighted naïve bayes and support vector machines [41, 43].

Table 1. When to use filters, wrappers and embedded feature selection methods.

Method	Measures	Search	Assessment	Decision Criteria
Filter	Feature subset relevance base on intrinsic properties of features	Usually order features	Uses statistical tests	Pre-processing technique. Fast processing speed and robust against overfitting but may fail to select the most useful features.
Wrapper	Feature usefulness based on classifier performance	Search the space of all feature subsets	Uses classification cross-validation performance	Finds the most useful features, but is prone to overfitting. Better for non-linear class separation.
Embedded	Feature subset usefulness	Search guided by learning process	Uses cross-validation	Like wrappers but less computationally expensive and less prone to overfitting.

2.3. Reducing Dimensionality: Feature Extraction Techniques

The key difference between feature selection and feature extraction is that feature selection keeps a useful subset of the original features while feature extraction creates brand new ones. Feature extraction methods transform the data from higher to lower dimensions. This paper will touch on three of the most popular techniques.

2.3.1. Feature Extraction Techniques – Principal Component Analysis

The central idea of principal component analysis (PCA) is to reduce the dimensionality of a data set consisting of numerous interrelated variables, while retaining as much as possible of the variation present in the data set [44]. PCA transforms a set of potentially correlated variables into a set of uncorrelated linear combinations of those variables called principal components [44], by projecting the original data in the direction where the variance between data points is maximized. The resulting transformation is a two-dimensional scatter plot making data easier to visualize.

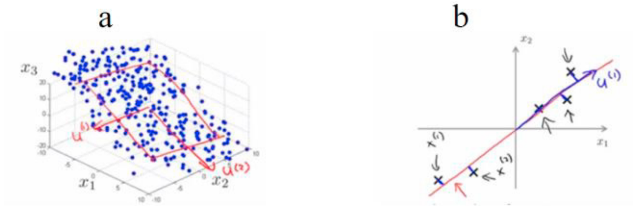

Fig. 1. (a) Data in three-dimensional space before applying PCA (b) Data in two-dimensional scatter plot after applying PCA [62]

The weights of the principal components are derived from the eigenvectors of the co-variance or correlation matrix from the data set [45]. The first principal component is a straight line that minimizes the sum of the squared distances from the data points. Therefore, the first principal component explains the highest amount of variance from the data set [45]. The residuals are then extracted from the data-set and the next principal component is calculated. As such, each successive component explains less of the variance. It is often sufficient to select the first principal components, where the amount of variance explained by the principal components is greater than or equal to some threshold, thereby reducing the dimensionality of the problem from the number of variables, to the number of principal components [45]. As an unsupervised learning method, PCA does not consider outcome or response variables.

Some challenges exist when using PCA [45]. First, the algorithm is sensitive to the scale of the variables in the data set [45]. Therefore, it is advisable to perform mean centering and use the correlation matrix, as it is normalized. Failure to normalize the data would cause the features on the largest scale to dominate the new principal components. Also, PCA is not optimal for classification purposes. Finally, a threshold for cumulative explained variance must be set or tuned.

2.3.2. Linear Discriminant Analysis

Linear Discriminant Analysis (LDA) is a method of dimensionality reduction that attempts to find a linear combination of variables to categorize or separate two or more groups [46]. Like PCA, LDA creates linear combinations of the original features. However, unlike PCA, LDA doesn't maximize explained variance of the attributes. Instead, it maximizes the separation between classes of attributes. LDA works when the measurements made on independent variables for each observation are continuous quantities. When dealing with categorical independent variables, the appropriate technique is discriminant correspondence analysis [47].

Therefore, LDA can only be used with labeled data. As a supervised learning technique, a properly defined training set must be provided for the model to learn [47]. For example, comparisons between classification accuracies for image recognition after using PCA or LDA show that PCA tends to outperform LDA if the number of samples per class is relatively small [48]. In practice, it is not uncommon to use both LDA and PCA in combination. For example, PCA can be used first for dimensionality reduction followed by LDA for classification [49]. Alternatively, LDA can be used first to obtain a cluster-preserved low-dimensional data set, and PCA is then applied to further reduce the dimensions to two for visualization [50].

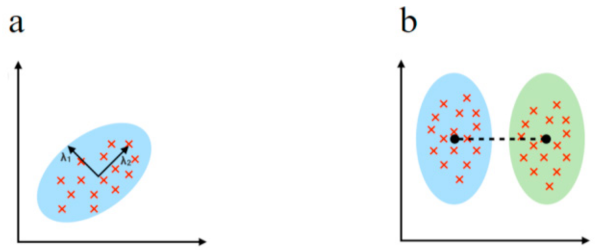

Fig. 2. (a) PCA finds the axes that maximize attribute variance (b) LDE finds axes that maximize class separation [49]

2.3.3. T-SNE

T-distributed stochastic neighbor embedding (t-SNE) is a dimensionality reduction method well-suited for embedding high-dimensional data for visualization in a low-dimensional space [51]. Specifically, it models each high-dimensional object by a two or three-dimensional point in such a way that similar objects are modeled by nearby points and dissimilar objects are modeled by distant points with high probability [51]. The t-SNE algorithm comprises of two main stages. First, t-SNE constructs a probability distribution over pairs of high-dimensional objects so that similar points have a higher probability of being picked, while dissimilar points have a smaller probability of being picked [51]. Second, t-SNE defines a similar probability distribution over the points in the low-dimensional map, and it minimizes the Kullback–Leibler divergence between the two distributions with respect to the locations of the points in the map [51]. Kullback–Leibler divergence is a measure of how one probability distribution is different from a second, reference probability distribution.

T-SNE differs from the methods listed above in that t-SNE is a non-linear method and performs better on data where the underlying relationship is not linear [51]. This method creates a 2D plane or 3D manifold that preserves the relative distance between observations. This output is then easily visualized as a scatterplot, aiding in visualization [51]. T-SNE works quite well where other methods struggle, and can often find structure where other dimensionality reduction algorithms cannot. Unfortunately, that flexibility makes it tricky to interpret [52].

T-SNE is not a clustering algorithm, but rather a dimensionality reduction algorithm. Because it maps multi-dimensional data into a lower dimensional space, the input features are no longer identifiable after transformation. Thus, you cannot make any inference based solely on the output of t-SNE, which is why it's use cases are mainly data exploration and visualization. However, t-SNE can be used in the process of classification and clustering by using its output as the input feature for other algorithms [53]. The main problem while using t-SNE is the black box type nature of the algorithm. Another problem with the algorithm is that it doesn't always provide a similar output on successive runs. Therefore, the best ways to use the algorithm is for exploratory data analysis or as an input parameter for other classification and clustering algorithms [7].

Table 2. When to use t-SNE, PCA and LDA.

Method	Algorithm Type	Data Structure	Decision Criteria
LDA	Supervised	Linear	Ideal for classification of multiple groups
PCA	Unsupervised	Linear	Ideal for initial data exploration
t-SNE	Supervised	Non-linear	Ideal for exploring and visualizing high-dimensional data and as an input parameter to other classification & clustering algorithms

3. Numerosity Reduction Techniques

Unlike dimensionality reduction, the goal of numerosity reduction is not to reduce the number of dimensions or variables in the data, but instead to compress the volume of data by choosing or creating smaller forms of data representation [54]. Techniques include: (i) parametric models, which store only the model parameters instead of the actual data, and (ii) nonparametric methods, such as clustering, sampling, and the use of histograms, which store representations of the data [55].

3.1. Reducing Numerosity: Parametric Methods

For parametric methods, a model is used to estimate the data, so only the data parameters need to be stored instead of the actual data [55]. Regression and log-linear models can be used to approximate the given data. In simple linear regression, the data are modeled to fit a straight line [55]. Multiple linear regression is an extension of simple linear regression, which allows a response variable to be modeled as a linear function of two or more predictor variables.

Log-linear models approximate discrete multidimensional probability distributions. Log-linear models can be used to estimate the probability of each point in a multidimensional space for a set of discretized attributes, based on a smaller subset of dimensional combinations [55]. This allows a higher-dimensional data space to be constructed from lower dimensional spaces. Regression and log-linear models can both be used on sparse data, although their application may be limited. While both methods can handle skewed data, regression does so exceptionally well. Regression can be computationally intensive when applied to high dimensional data, whereas log-linear models show good scalability for up to ten dimensions [55].

3.2. Reducing Numerosity: Non-Parametric Methods

Nonparametric methods for storing reduced representations of the data include histograms, clustering, and sampling [55]. Histograms use binning to approximate data distributions and are a popular form of data reduction. A histogram partitions the data distributions into buckets, often transforming quantitative information into qualitative information.

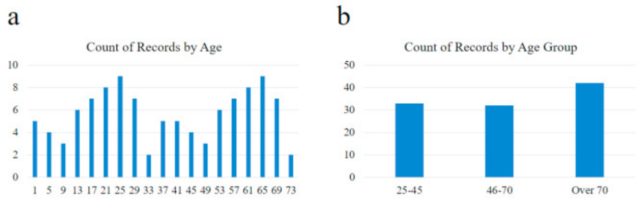

Fig. 3. (a) Data before histogram/binning; (b) Data after histogram/binning

Numerous partitioning algorithms can be applied to define range of values [56]. A clustering algorithm can also be applied to partition data into clusters or groups based on similarities. Each cluster forms a node of a concept

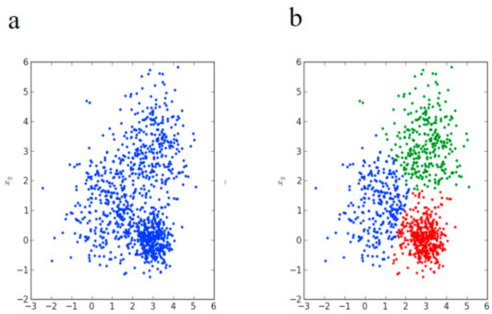

Fig. 4. (a) Original data before k-means clustering algorithm; (b) Data after k-means clustering algorithm

hierarchy, where all nodes are at the same conceptual level. Each cluster may be further decomposed into sub-clusters, forming a lower level in the hierarchy. Clusters may also be grouped together to form a higher-level concept hierarchy [54]. Clustering algorithms include connectivity-based clustering (hierarchical clustering), centroid-based clustering, distribution-based clustering and density-based clustering [54]. After clustering algorithms are applied, data can be aggregated for analysis according to the centroid within each cluster (most representative point). If the centroid must be a specific data point instead of a midpoint between the data, then the median may be used instead of the mean.

With sampling, only a small subset of the data is returned to represent the data set as a whole. The goal is to find a statistically representative sample. There are two main types of sampling: probability (representative) and non-probability (non-representative) [57]. With representative sampling, everyone in the population has an equal likelihood of selection in the sample. Alternatively, with non-probability sampling, everyone in the population does not have an equal likelihood of selection. Non-probability sampling can be intentional such as when a quota by certain demographics are desired. Probability sampling techniques include: simple random, systematic random, stratified random, cluster, and multi-stage sampling [58]. Non-probability sampling techniques include: convenience/purposive, quota, judgment, and snowball sampling [58].

3.2.1. Discretization and Concept Hierarchy Generation

Data discretization reduces the number of values for a given continuous attribute by dividing the range of continuous numeric attributes into intervals [55]. In this manner, symbolic data mining algorithms can be applied over continuous data and the representation of information is simplified, making it more concise and specific. Interval labels can then be used to replace actual data values. Histograms are a form of data discretization.

Concept hierarchies reduce the data through aggregation, by collecting and replacing low level concepts (such as numeric values for the attribute age) by higher level concepts (such as young, middle-aged, or senior). Although detail is lost by such data generalization, the generalized data may be more meaningful and easier to interpret [59]. The smaller the number of distinct values to sort, the faster these methods should be. Concept hierarchies are ideal for rolling up geographic data such as zip code → city → county → state → country →continent.

Data discretization and concept hierarchies aim to aggregate lower levels of data to the highest level of data necessary to perform efficient analysis and visualization. With this goal in mind, data architecture also plays a key role in optimizing the visualization experience.

4. Conclusion

The return on investment (ROI) of big data visualization is well-documented in numerous studies and use cases [60, 61]. However, to achieve ROI from analytics investments, key insights must be uncovered, understood and communicated. Synthesizing huge quantities of data into key insights grows more challenging as data volumes and varieties increase. To address visualization challenges posed by big and high dimensional data, this paper explores algorithms and techniques that compress the amount of data and/or reduce the number of attributes to be analyzed and visualized. By combining these approaches several benefits result. First, the storage space required for computing in-memory visualization is reduced and rendering speeds increase. Next, computational resources and consumption based cloud data costs are decreased, as there is less data to process. Also, model error and accuracy are improved. Finally, it becomes easier to visualize patterns and trends more clearly, allowing key insights to be generated.

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